INDUSTRIAL HAZARDS AND SAFETY

DEFINITIONS:

- HAZARD: A substance that causes an injury in a given environment or situation.
- INDUSTRIAL HAZARD: Any condition produced by industries that may cause injury or death to personnel or loss of product or property.
- SAFETY: Freedom from the occurrence of risk or injury or loss.
- INDUSTRIAL SAFETY: Protection of workers from the danger of industrial accidents.
- **Toxic and corrosive chemicals, fire, explosions and personnel falling-causes of accidents.**

CAUSES OF ACCIDENTS:

- Human failure:
  - Operating without clearance, failure to respond warning
  - Working at an unsafe speed
  - Using equipment unsafely
  - Making safety devices un-operative
  - Unsafe loading, placing, mixing
  - Distracting, teasing, abusing

Management of over exposure to chemicals:

1. Removal from exposure
2. Resuscitation: restoration of life
3. Decontamination
4. Symptomatic treatment: convulsions, bronchospasms, dehydration, and arrhythmias.

What are the effects???

- From irritation to death
- Severe burns by destroying living tissues- dehydration, digestion, oxidation
- Eyes, nervous system, nose, mouth and throat, pulmonary system, CVS, Breast, digestive system, genitourinary system, skin

- Dermatitis reactions to synthetic drugs- Acridines and phenothiazine compounds
- While grinding vegetable drugs dust evolved causes irritation- capsicum, podophyllum

- Compressed gases: filled in cylinders and transported to the place of use and the precautions are:
  - Cylinders should not be dropped or struck.
  - Safety devices should not be dropped.
  - Valves should be of high quality.
  - Protected against weather changes, especially excessive temperature.
  - Cylinders should bear a standard label indicating the kind of gas- inflammable, corrosive, or inert.
  - Full cylinders should be separated from empty cylinders.

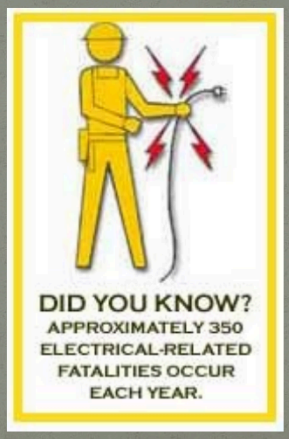
DUST EXPLOSION

- The term Dust is used if the Max particle size of the solids is 500 mm in the mixture.
- During grinding operations, stone or iron pieces results in sparks finally leads to explosion of combustible materials.
- Dust of starch, dextrin and some organic materials are dangerous.
- Mixture of combustible dust burns or oxidizes in presence of air.
- Upper and lower limits of air and dust are given for the explosion to takes place.

Methods to exclude dust explosion

- Avoiding the development of explosive mixtures
- Replacing the atmospheric oxygen with inert gas, working in a vacuum, or using inert dust-rock *salt and sod sulfate*
- Preventing the occurrence of an effective ignition source- *welding, smoking, cutting, mechanically generated sparks*

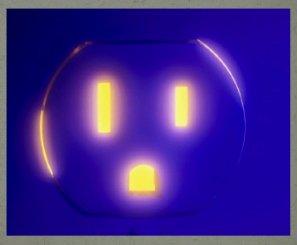
Electrical Hazards



Why know about it?

Because failing to take the necessary precautions can lead to:

- injury or death
- fire or property damage



What are the types of injuries?

- Burns
- Shocks
- Falls



KEEP SAFE

KNOW HOW TO CONTROL ELECTRICAL HAZARDS

How do you respond to electrical incidents?

If you come across a person receiving an electric shock:

- if possible, disconnect the electrical supply (switch?)
- assess the situation – never put yourself at risk
- take precautions to protect yourself and anyone else in the vicinity
- apply the first aid principles
- assess the injuries and move the casualty to a safe area if required
- administer first aid if trained
- seek urgent medical attention



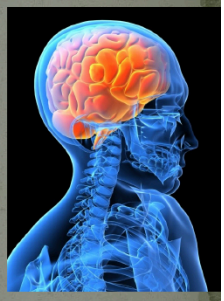
Do you want to be a victim?

You could be the victim if you:

- don't follow proper procedures around electricity
- use electrical equipment improperly
- use faulty electrical equipment

What are the levels of effect of current?

AC current (mA)	Effect on human body
1	Slight tingling sensation
2-9	Small shock
10-24	Muscles contract causing you to freeze
25-74	Respiratory muscles can become paralysed; pain; exit burns often visible
75-300	Usually fatal; ventricular fibrillation; entry & exit wounds visible
>300	Death almost certain; if survive will have badly burnt organs and probably require amputations



Accident prevention & reduction

Treatment



SAFETY MANAGEMENT

- Types of accidents:
  - Major results in death or prolonged disability.
  - Minor scratch or a cut that doesn't seriously disable the worker.

Advantages of safety management:

1. Cost saving: Direct costs-compensation medical expenses

   Indirect costs to 4 times higher include Loss due to downtime operations, slowed-up production rate, spoiled materials, labor time, charges for cleaning, and damage of equipment.

2. Increased productivity: employees can devote more time in improving the quality and quantity of output.
3. Moral:boosts morality of the worker and his family.
4. Legal: laws covering occupational health and safety and the penalties for noncompliance are severe.

- To make supervision fully accountable for safety performance in the working areas.
- To train in safe methods of working and provide education and guidance on prevention of hazards.

ELEMENTS OF THE SAFETY PROGRAMME:

- Strategic choice
- Safety policy
- Organization for safety
- Extent and remedies for accidents
- Training in safety
- Safety program evaluation

- Strategic choice: To make decisions regarding safety of workers.
  As a marketing tool.
- Safety policy: must contain a declaration of 4 fundamental points-

i. Safety of employees and the public is paramount importance.
ii. Safety will take precedence over expediency.
iii. In the development and implementation of safety procedures every effort will be made to involve all managers, supervisors and employees.
iv. Safety legislation will be complied in letter and the spirit of the law.

- safety organization:

Constitution of an organization- Safety committee.

composed of employees.

serves as advisory capacities

responsible for – reviewing safety procedures.

-making recommendations for eliminating health hazards.

-investigating accidents

-fielding safety-related complaints from employees.